

AI-Powered Air Quality Forecasting

Delhi AirCast

*Hyperlocal Short-Term PM_{2.5} and AQI Forecasting
for Individual Monitoring Stations in Delhi*

Data Sources, Acquisition Architecture and System Design Report

*Free Indian government and weather data, checked and ready to use,
for a neighbourhood air-quality forecast in Delhi.*

Undergraduate Technical Project Report

City / study region	Delhi, National Capital Territory of India
Primary target	Hourly PM _{2.5} (µg/m ³), converted to Indian National AQI
Forecast horizon	1–24 hours ahead, neighbourhood / station granularity
Verification date	7 September 2026
Status	Verified system design · 7 September 2026

Submitted in partial fulfilment of the project requirement

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Project guide

Date

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*All sources used in this report are free for academic use.
Live endpoints were probed on 7 September 2026 before the design was frozen.*

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Note. This document is the data, architecture and modelling plan for Delhi AirCast. We build the project from this plan.

1. Introduction

People in Delhi need air information they can use: their own neighbourhood, and the next few hours. That helps someone decide whether to run at 6 a.m., whether a school playground stays open at 4 p.m., or how the next six hours look in their area. Problem Statement PS-13 asks for an AI system that forecasts air quality at neighbourhood scale using station readings, weather, and links between nearby monitoring stations.

We start with good data, then we add the AI. This report shows the free CPCB and weather sources we checked on 7 September 2026, and the clean hourly table we build from them for Delhi.

The system is named Delhi AirCast. It predicts PM2.5 and the Indian National Air Quality Index 1–24 hours ahead for individual monitoring locations, using historical pollution, meteorology, time features and, in a later phase, neighbouring-station information.

2. Problem Statement and Objectives

2.1 Formal problem statement

PS-13 — AI-Powered Air Quality Forecasting. Citizens lack hyperlocal, short-term air quality forecasts to plan outdoor activity, especially in polluted cities. The suggested solution is a time-series / graph neural network model that combines sensor and weather data to forecast AQI at neighbourhood granularity. Suggested technology includes LSTM / Graph Neural Network, CPCB sensor data, a weather API, and an interactive map dashboard.

2.2 Objectives of this report

1. Identify every usable free air-quality and weather source relevant to Delhi.
2. Distinguish ground-station measurements from satellite / chemical-transport model fields.
3. Check which free endpoints respond quickly enough for a college project.
4. Set a clear job for each source: training data, live updates, weather, and extra background.
5. Define the training table, live update cadence, AQI conversion, modelling stages and dashboard.
6. Lay out the model path: simple baseline, then XGBoost, then LSTM, then a station graph.

3. Scope, Assumptions and Design Principles

3.1 Scope

- **Geography.** Delhi (NCT) for Phase 1. The same method can later cover Pune and other cities.
- **Stations.** Eight to ten CPCB/DPCC/IMD/IITM stations spanning east, west, south, north and the airport corridor.
- **Horizon.** Forecasts 1, 6, 12 and 24 hours ahead.
- **Resolution.** Hourly. 15-minute CPCB files are averaged to the hour for modelling.
- **Target.** The model predicts PM2.5. The app converts that number to Indian AQI.
- **Cost.** Free academic sources: OpenCity, data.gov.in, OpenAQ and Open-Meteo.

3.2 Design principles

- **Real station data.** We train on CPCB station readings and show live CPCB updates on the dashboard.
- **Fast weather.** Weather comes from Open-Meteo, which is free and works in under a second.

- **Step by step.** We start with a simple baseline, then XGBoost, then LSTM, then a station graph. Each step has to improve the last one.
- **Clear method.** The model learns patterns from past pollution, weather, time of day and nearby stations.

4. What We Are Building

Given current and recent pollution plus weather at one Delhi station, forecast PM2.5 for the next few hours, convert that forecast to Indian AQI, and tell a citizen whether outdoor air is expected to improve or deteriorate. A concrete example:

At 10:00 at Anand Vihar: PM2.5 = 145 $\mu\text{g}/\text{m}^3$, PM10 = 230, temperature = 27 °C, humidity = 72 %, wind = 4 km/h, no rain. The model emits a six-hour PM2.5 path (for example 151, 158, 163, 157, 149, 142), an AQI near 190 (Poor), and a sentence: “Air quality is expected to deteriorate over the next three hours.”

That is already a legitimate AI forecasting system. Neighbourhood granularity comes from modelling stations separately (and, later, as a graph). Short-term comes from the 1–24 h horizon. The map dashboard is the public interface. Every week we add new data and keep the better model.

5. Verified Data Landscape

We use three kinds of free data, and each has a clear job.

Table 1. Three classes of free data, and the role of each in Delhi AirCast.

Class	What it actually is	Use in Delhi AirCast
Ground station (CPCB / DPCC / IMD / IITM)	A physical analyser at a named site, typically 15-min or hourly PM2.5, PM10, NO2, SO2, CO, O3, plus on-site meteorology.	PRIMARY label and live current conditions
Meteorological reanalysis / forecast	Gridded weather (ERA5 / ECMWF IFS) interpolated to each station.	PRIMARY weather features and future-hour weather
Chemical transport / CAMS model	Satellite-assimilated regional air quality at about 45 km, smooth and complete.	EXTRA background feature and a useful comparison

Table 2 summarises every source that was either downloaded or called on 7 September 2026. Latency numbers are wall-clock times from this workstation on 7 September 2026.

Table 2. Source scorecard after live verification on 7 September 2026.

Verdict	Source	Free?	Key?	What we got	Latency
USE — primary train	OpenCity CPCB dumps	Yes, public domain	No	Anand Vihar 15-min, 2024-01-01 to 2025-12-31, 70,177 rows, full pollutants + station meteo	3.4 s / 11.8 MB
USE — live official	data.gov.in CPCB API	Yes (NDSAP)	Free key	Delhi stations updating 07-09-2026 21:00 IST (Narela, Rohini, Anand Vihar, NSIT Dwarka, ...)	~3 s
USE — weather	Open-Meteo forecast + archive	Yes	No	Delhi hourly temp/humidity/wind/rain/pressure;	0.5–0.7 s

				archive week of 2024-01-01 returned 168 hours	
USE — hist/live backup	OpenAQ v3	Yes	Free key	Free key needed. CPCB Anand Vihar is location 235, reporting since 2016	ready with key
EXTRA feature	Open-Meteo Air Quality (CAMS)	Yes	No	Regional background PM2.5 for Delhi, 21:30 IST = 103.1 $\mu\text{g}/\text{m}^3$	0.55 s
EXTRA live	WAQI / aqicn	Yes	Free token	Third live feed once we have a personal token	ready
EXTRA history	OpenAQ + Vonder CPCB archives	Yes	Free key / none	More years and more stations when we want a longer record	ready

6. Layer A — Primary Historical Air Quality Data

6.1 OpenCity CPCB / DPCC station files (recommended core)

The strongest training source found is the OpenCity dataset “Delhi Hourly Air Quality Reports”. It republishes CPCB Continuous Ambient Air Quality Monitoring Station (CAAQMS) files for 39 Delhi sites. The 2024–2025 series is 15-minute resolution with pollutant concentrations and on-site meteorology. Licence is listed as public domain. The original source cited is the CPCB data repository. We use that official Indian monitoring data through a fast public copy that is easy to download.

Direct package API: https://data.opencity.in/api/3/action/package_show?id=delhi-hourly-air-quality-reports (HTTP 200 in 0.19 s; 78 resources). The Anand Vihar 2024–25 file was downloaded in full and profiled.

Table 3. Profile of the Anand Vihar 2024–2025 OpenCity extract.

Field	Observed value (Anand Vihar, site_301)
File	del-anand-vihar-dpcc-2024-25.csv (11.8 MB)
Rows	70,177 (15-minute)
Time span	2024-01-01 00:00 UTC → 2025-12-31 23:45 UTC
Pollutants	PM2.5, PM10, NO, NO2, NOx, NH3, SO2, CO, Ozone, BTX
Station meteorology	AT (°C), RH (%), WS (m/s), WD (deg), RF (mm), BP, SR
PM2.5 missing	10,626 rows (15.1 %) — usual station gaps; we mark them and keep going
PM2.5 range	1.0 – 998 $\mu\text{g}/\text{m}^3$ (998 is a known sensor spike; we clean it in QC)
Licence	Public Domain (OpenCity metadata)

Ten stations hourly-aggregated over two years give on the order of $10 \times 24 \times 365 \times 2 \approx 175,000$ station-hours. That is a complete training set. Two full years already include winter, monsoon, post-monsoon and Diwali. We use the 2024–2025 15-minute files because they have the full pollutant columns and a timestamp on every row.

6.2 How we get CPCB history quickly

The official CAAQMS files come from CPCB. OpenCity, OpenAQ and the Vonder parquet releases already publish those same station readings in an easy, citable form. CPCB is the origin; OpenCity and OpenAQ are the fast access path we use.

7. Layer B — Official Live CPCB Snapshot

India's Open Government Data platform publishes "Real Time Air Quality Index from various locations" (resource ID 3b01bcb8-0b14-4abf-b6f2-c1bfd384ba69), contributed by CPCB under the Ministry of Environment, Forest and Climate Change. A sample-key call with city = Delhi and pollutant = PM2.5 returned HTTP 200 and stations whose last_update was 07-09-2026 21:00:00 — the same evening this report was written. Example stations in that payload: Narela, Rohini, Sri Aurobindo Marg, Najafgarh, Talkatora Garden, NSIT Dwarka, Anand Vihar, Vivek Vihar.

This endpoint is the official live snapshot for the dashboard. A personal data.gov.in key lets us pull every Delhi station and every pollutant in one job. Pollutant identifiers observed in community documentation: PM2.5, PM10, NO2, NH3, SO2, CO, OZONE.

Field names in the JSON are min_value / avg_value / max_value. We train on OpenCity / OpenAQ concentrations in $\mu\text{g}/\text{m}^3$. On the live dashboard we show the official CPCB number clearly, and we check it against the same hour in OpenCity so the units stay clear.

[https://api.data.gov.in/resource/3b01bcb8-0b14-4abf-b6f2-c1bfd384ba69?api-key=YOUR_KEY&format=json&filters\[city\]=Delhi&limit=1000](https://api.data.gov.in/resource/3b01bcb8-0b14-4abf-b6f2-c1bfd384ba69?api-key=YOUR_KEY&format=json&filters[city]=Delhi&limit=1000)

8. Layer C — OpenAQ Archive and Live Backup

OpenAQ aggregates reference-grade monitors worldwide and exposes them through a documented REST API (v3). India CPCB stations, including Anand Vihar, New Delhi – DPCC, are present (OpenAQ location 235; reporting since 2016; parameters include PM2.5, PM10, NO2, SO2, CO, O3 and some on-site meteorology). A free key from explore.openaq.org unlocks the full API.

OpenAQ gives us extra years before 2024, a second live path, and concentrations in $\mu\text{g}/\text{m}^3$. It is the second free key in the stack, after data.gov.in.

9. Layer D — Weather: Open-Meteo

Weather is the easy part, and it should stay easy. Open-Meteo requires no API key for ordinary academic use. Two endpoints were verified:

Table 4. Open-Meteo weather probes (7 September 2026).

Endpoint	Query	Result
api.open-meteo.com/v1/forecast	Delhi 28.6139, 77.2090; hourly temp, RH, rain, pressure, wind, cloud; past 2 days + 1–2 day forecast; Asia/Kolkata	HTTP 200 in 0.53 s; 72-hour series, first 2026-09-05 00:00, last 2026-09-07 23:00
archive-api.open-meteo.com/v1/archive	Same point, 2024-01-01 to 2024-01-07	HTTP 200 in 0.74 s; 168 hours; temperature at 00:00 IST = 7.0 °C

OpenCity files already include station weather, and Open-Meteo adds a complete history plus the next day's forecast. That gives the model tomorrow's wind, rain and temperature while it predicts the next hours of PM2.5 from recent readings.

10. Layer E — Auxiliary Free Sources

10.1 Open-Meteo Air Quality (CAMs) — extra regional background

Open-Meteo also serves Copernicus Atmosphere Monitoring Service (CAMs) global composition. The global domain is about 0.4° (~ 45 km), updated twice daily, with a multi-day forecast. A live call for Delhi returned $\text{PM}_{2.5} = 103.1 \mu\text{g}/\text{m}^3$, $\text{PM}_{10} = 113.9$, $\text{US AQI} = 159$ at 2026-09-07 21:30. A 2024-01-01 week of history returned 168 hourly $\text{PM}_{2.5}$ values (example 146.7, 130.4, 117.8). We use this as extra regional background, as a second comparison, and as a helpful fill when a station is briefly down. The station forecast still comes from CPCB readings.

10.2 WAQI / World Air Quality Index

WAQI republishes many government stations and offers a free non-commercial token. A personal token gives us an extra live feed for the dashboard.

10.3 Daily CPCB AQI bulletins (UrbanEmissions)

UrbanEmissions.info maintains processed CPCB daily AQI bulletins for Indian cities (2015–2024). We use these city-day numbers for the exploration chapter: winter vs monsoon, Diwali spikes, and year-to-year drift of city AQI. GitHub: [urbanemissionsinfo/AQI_bulletins](https://github.com/urbanemissionsinfo/AQI_bulletins).

10.4 Vonter india-cpcb-aqi parquet

The GitHub project Vonter/india-cpcb-aqi publishes 15-minute national CAAQMS extracts and hourly AQI parquet files (latest release tag 2025). If OpenCity is missing a Delhi station-year, this is another free CPCB archive we can add.

10.5 Vayuayan

The vayuayan Python package wraps CPCB historical and live clients. We can use it later if we want another way to pull CPCB files.

11. Other Public Datasets

Many public Delhi files exist on Hugging Face, Kaggle and Zenodo. They are useful for extra reading and for later experiments. For training and for the live app we stay with OpenCity CPCB station files, the official data.gov.in feed, OpenAQ, and Open-Meteo, because we have already opened those files and confirmed the hours, stations and units.

12. Recommended Multi-Source Architecture

Delhi AirCast uses several free sources, each with a clear job: station readings for training, weather for features, the official API for live “now”, and extra feeds as backup.

Table 5. Frozen multi-layer free-data architecture.

Layer	Source	Cadence	Role
A Historical labels	OpenCity CPCB 2024–2025 (8–10 Delhi stations)	One-time bulk + optional yearly refresh	Train / validate $\text{PM}_{2.5}$
A2 Extra years	OpenAQ CPCB measurements; Vonter parquet if needed	One-time back-fill	2018–2023 optional

B Live official	data.gov.in CPCB resource 3b01bcb8-...	Every hour	Dashboard “now”
C Live / hist backup	OpenAQ v3 locations and measurements	Every hour / on B failure	Backup now + archive
D Weather hist+fcst	Open-Meteo archive + forecast	Hourly; 48 h forecast	Features now and future
E1 Regional AQ model	Open-Meteo CAMS air-quality	Hourly	Auxiliary feature / baseline
E2 Optional live	WAQI personal token	Hourly if B and C fail	Tertiary nowcast
E3 City climate EDA	UrbanEmissions daily bulletins	Static	Report graphics only

Data flow in one sentence. OpenCity (and optionally OpenAQ) plus Open-Meteo history are cleaned and merged into a station-hour table; models are trained offline; every hour the live CPCB snapshot and Open-Meteo forecast are written to a database; the forecast engine emits PM2.5 paths; an AQI engine converts them; the web dashboard draws the map, charts and alerts.

CPCB/OpenAQ + Open-Meteo → clean & merge → train (XGBoost / LSTM [/ GNN]) → PM2.5 forecast → CPCB AQI → React (or similar) map dashboard. New measurements every hour. Retrain every week only if validation improves.

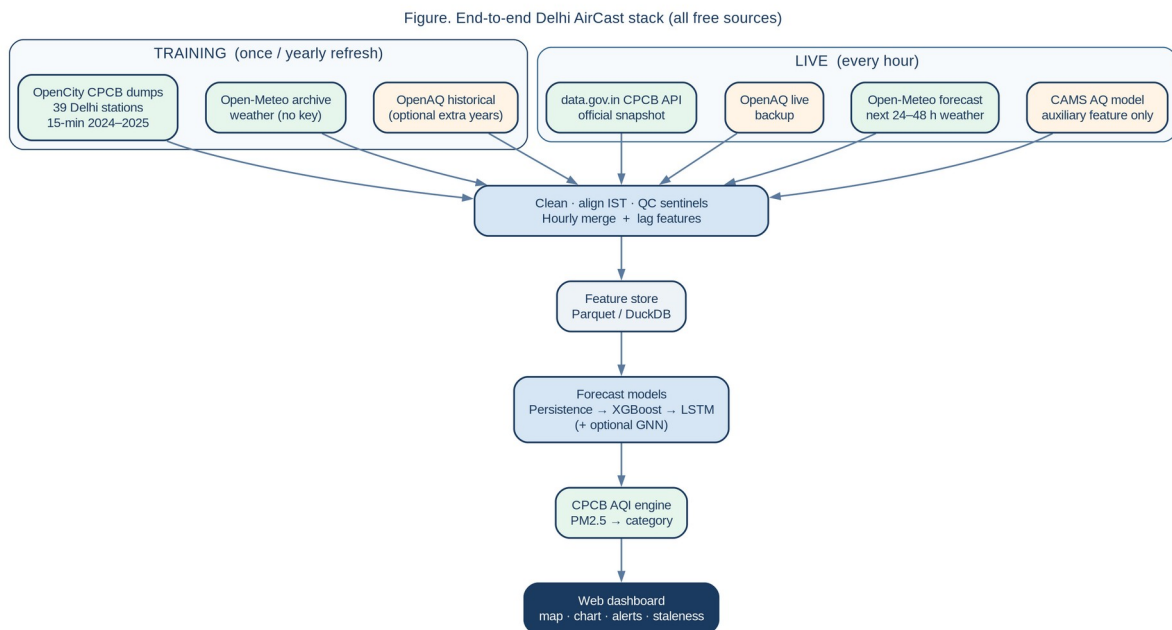


Figure 1. End-to-end Delhi AirCast stack. Green boxes are Phase-1 primary sources; amber boxes are backups or auxiliary features.

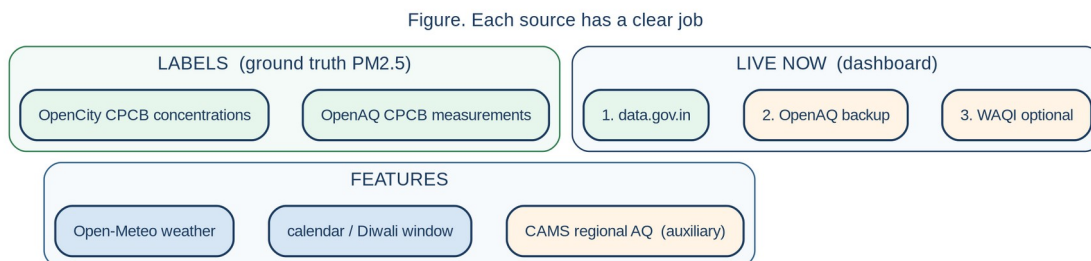


Figure 2. Training labels, weather features and live updates each have a clear job.

12.1 What already works today

On 7 September 2026 we opened the real files and APIs. OpenCity Anand Vihar came down in 3.4 seconds (11.8 MB). Open-Meteo weather and forecast replied in about half a second, with no key. The official data.gov.in CPCB feed was updating at 21:00 IST the same evening. OpenAQ and WAQI join the live path with free keys. This is a working stack we can build on.

13. Target Variable and Indian AQI Methodology

The model predicts hourly PM_{2.5} in $\mu\text{g}/\text{m}^3$. The app then converts that number to the Indian National AQI using CPCB's official 2014 table, so people see a familiar category such as Good, Poor or Severe.

CPCB computes a sub-index for each pollutant from its averaging period (24 h for PM_{2.5}/PM₁₀/NO₂/SO₂/NH₃, 8 h for CO and O₃) and takes the maximum sub-index as the location AQI, provided at least three pollutants are available and one of them is PM_{2.5} or PM₁₀, with at least 16 hours of data. Categories are Good (0–50), Satisfactory (51–100), Moderately Polluted (101–200), Poor (201–300), Very Poor (301–400), Severe (401–500). PM_{2.5} breakpoints are 0–30, 31–60, 61–90, 91–120, 121–250, 250+ $\mu\text{g}/\text{m}^3$. The full table is reproduced in Appendix C. The AQI engine follows CPCB's published Indian method.

The dashboard reports three numbers for each forecast hour: (i) hourly PM_{2.5}, (ii) a rolling 24 h mean PM_{2.5} converted to a CPCB-style sub-index so that the number is comparable to Sameer, and (iii) the plain-language category. The averaging window is printed on the card.

14. Dataset Schema and Feature Engineering

14.1 Canonical station-hour table

After resampling 15-minute OpenCity files to the hour (mean for concentrations and meteorology, sum for rainfall) and joining Open-Meteo on timestamp (Asia/Kolkata), each row looks like:

Table 6. Canonical merged schema (one row = one station-hour).

Column	Origin	Notes
timestamp_ist	OpenCity / OpenAQ	Hour beginning, Asia/Kolkata
station_id	CPCB site_*	e.g. site_301
station_name	CPCB	e.g. Anand Vihar, Delhi - DPCC
latitude, longitude	CPCB / OpenAQ	Fixed per station
pm25, pm10, no2, so2, co, o3, nh3	OpenCity / OpenAQ	$\mu\text{g}/\text{m}^3$ except CO in mg/m^3 as reported
temp_station, rh_station, wind_station, ...	OpenCity	Station weather, used together with Open-Meteo
temp, rh, wind_speed, wind_dir, pressure, rain, cloud	Open-Meteo	Complete series + forecast
cams_pm25, cams_pm10, cams_no2	Open-Meteo AQ	Optional regional features
hour, dow, month, season, is_weekend	calendar	Cyclical encodings for hour/month
is_diwali_window, is_crop_burning	calendar rules	Festival and crop-burning season flags

14.2 Lag and horizon features

For a forecast issued at time t , the model uses PM_{2.5} at t , $t-1$, $t-2$, $t-3$, $t-6$, $t-12$, $t-24$ and the same lags for wind, humidity and rain; plus hour-of-day, day-of-week, month and season; plus Open-Meteo weather

from t to $t+h$. Multi-horizon heads ($h = 1, 6, 12, 24$) are trained either as separate models or as a multi-output model. A graph model also uses recent PM_{2.5} at neighbouring stations.

14.3 Phase-1 station set

Ten stations that cover Delhi's spatial story and appear in both the OpenCity 2024–25 dump and the live data.gov.in feed:

Table 7. Phase-1 hyperlocal station panel.

Station	Operator	Why it is in the set
Anand Vihar	DPCC	East Delhi traffic / ISBT; consistently high PM; problem-statement example
Dwarka Sector 8	DPCC	South-west residential contrast
NSIT Dwarka	CPCB	Second Dwarka node; useful for a mini-graph later
IGI Airport T3	IMD	Airport corridor, different siting
Okhla Phase-2	DPCC	South-east industrial / residential mix
Rohini	DPCC	North-west; winter inversion stories
Punjabi Bagh	DPCC	West Delhi; long CPCB record
R K Puram	DPCC	South-central; often in media AQI tables
ITO	CPCB	Central traffic corridor
Mandir Marg	DPCC	Central / ridge-adjacent contrast

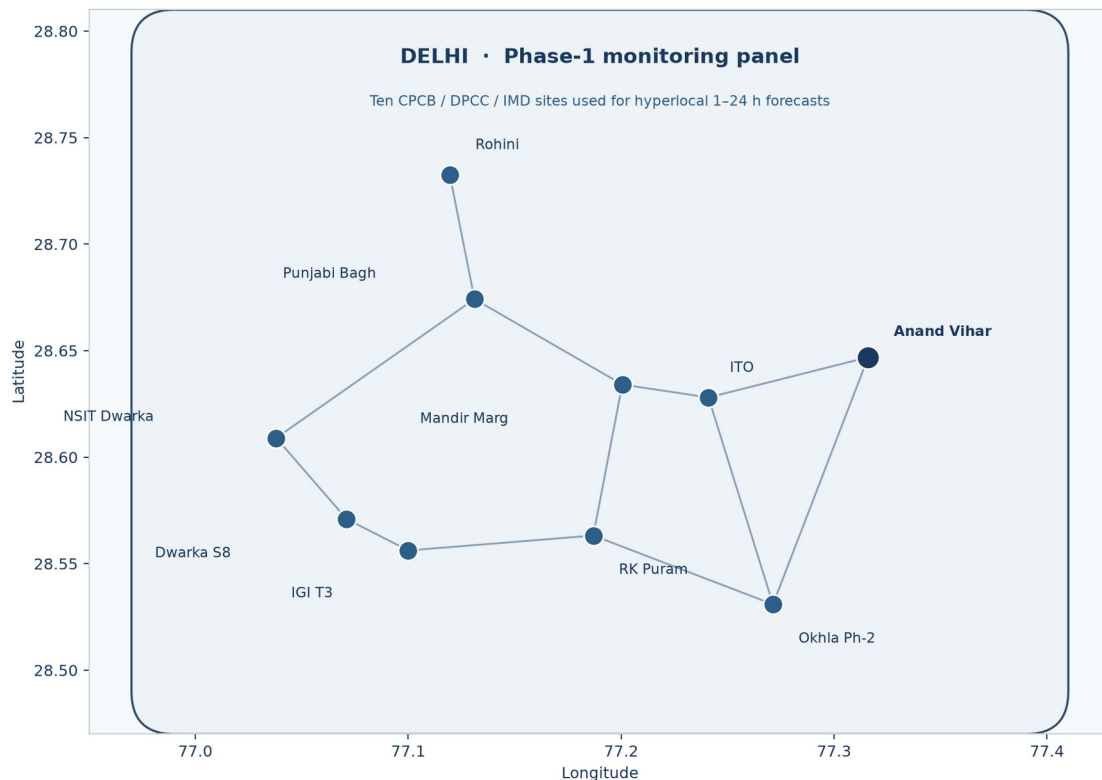


Figure 3. Phase-1 station panel on approximate WGS84 coordinates. Lines show neighbourhood links for a later graph model.

15. Data Collection, Storage and Update Cadence

15.1 One-time historical build

7. Download the ten 2024–25 OpenCity 15-minute CSVs (CKAN URLs in Appendix A).
8. Parse timestamps to UTC then Asia/Kolkata; coerce numerics; map sentinels (999, 998, -9999, blanks) to NA.
9. Resample to hourly means (rainfall: sum). Require at least two valid 15-minute PM2.5 samples to keep an hour, else NA.
10. Pull Open-Meteo archive for each station coordinate, 2024-01-01 to yesterday.
11. Optionally pull Open-Meteo CAMS PM2.5/PM10/NO2 for the same window.
12. Left-join weather onto station-hours; store parquet partitioned by station and year.
13. Freeze a train window (e.g. 2024-01 to 2025-09) and a test window (e.g. 2025-10 to 2025-12, including Diwali).

15.2 Hourly live job

Every hour, in order: (1) data.gov.in Delhi snapshot → raw JSON; (2) if that fails or is stale > 2 h, OpenAQ latest measurements for the ten location IDs; (3) optional WAQI; (4) Open-Meteo forecast + current weather; (5) Open-Meteo CAMS current; (6) append to the live table; (7) run the forecast engine; (8) write predictions. Training happens in the weekly job.

15.3 Weekly learning job

Every week: append newly collected hours to the training store; score the production model on the new week; retrain a candidate; compare MAE / RMSE / skill vs persistence on a rolling 28-day window; and keep the better model. That is how the forecast keeps improving.

Figure. Two clocks: hourly forecast vs weekly improve-if-better

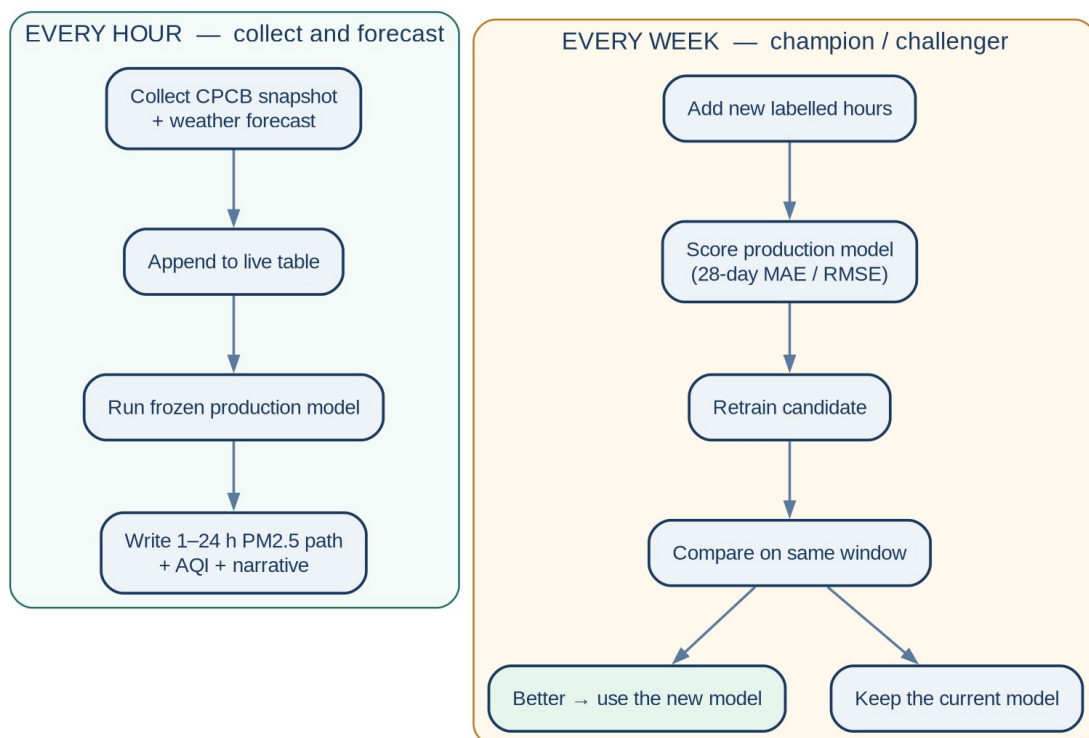


Figure 4. Two clocks. Every hour we collect data and forecast. Every week we add new data and keep the better model.

15.4 Storage

Phase 1 can be local parquet + SQLite or DuckDB (perfectly adequate for < 1 million station-hours). If a web app is deployed, Postgres is enough. Raw JSON payloads are kept for a 30-day audit so a bad parse can be replayed. Secrets live in a gitignored .env: DATA_GOV_IN_KEY, OPENAQ_API_KEY, optional WAQI_TOKEN.

16. Quality Control and Missing-Value Policy

- **Sentinels.** CPCB 999 / 998 / 985 / 1999 / negative concentrations → NA. Anand Vihar already showed a 998 $\mu\text{g}/\text{m}^3$ PM2.5 max.
- **Physical bounds.** Clip PM2.5 to [0, 1000] $\mu\text{g}/\text{m}^3$ after sentinel removal; log remaining extremes for Diwali / stubble-burning.
- **Clean training.** Lags may carry the last good reading for up to 3 hours. Training uses the real PM2.5 at that hour.
- **Gaps.** Hours with PM2.5 missing at t are left out of training for that horizon. CAMS stays an extra feature.
- **Weather.** When station weather is missing, we use Open-Meteo and keep the row.
- **Coverage.** Record completeness per station-month. Stations below 60 % completeness in a month get a quality flag on the map.
- **Units.** Unit check: CO in mg/m^3 vs $\mu\text{g}/\text{m}^3$; OpenAQ vs OpenCity vs data.gov.in sub-index.

17. Modelling Strategy

The problem statement mentions LSTM and Graph Neural Networks. We reach those after a simple baseline and an XGBoost model, so each step is a real improvement we can show.

17.1 Model 1 — Persistence baseline

$\hat{y}(t+h) = y(t)$ (last observed hourly PM2.5). A seasonal variant uses $y(t+h-24)$. This is the first score we beat. It is a fair, easy starting point and it already gives people a useful “same as now” forecast.

17.2 Model 2 — Gradient boosting (XGBoost or LightGBM)

Tabular model on lags, weather, calendar and optional CAMS background. This usually captures non-linear effects such as “wind falling + winter evening + no rain”. It is the workhorse and often the production model for a college timeline.

17.3 Model 3 — LSTM sequence model

Input: 24 (or 48) hours of multivariate history. Output: 1–24 h PM2.5 path. Trained per station or as a multi-station model with a station embedding. We keep it when it improves on boosting on the same split. This is the time-series deep learning step in PS-13.

17.4 Model 4 — Graph model (spatial rung)

Nodes = stations; edges = k-nearest neighbours by distance, or a distance kernel. A GNN (or a simpler spatial lag of neighbours’ PM2.5 inside XGBoost) tests whether Anand Vihar’s next hour is informed by Okhla, ITO and Vivek Vihar. This is how the project earns “neighbourhood granularity” beyond ten

independent point models. Neighbour lags inside XGBoost already encode most of that spatial story; the GNN is the same idea with learned edges.

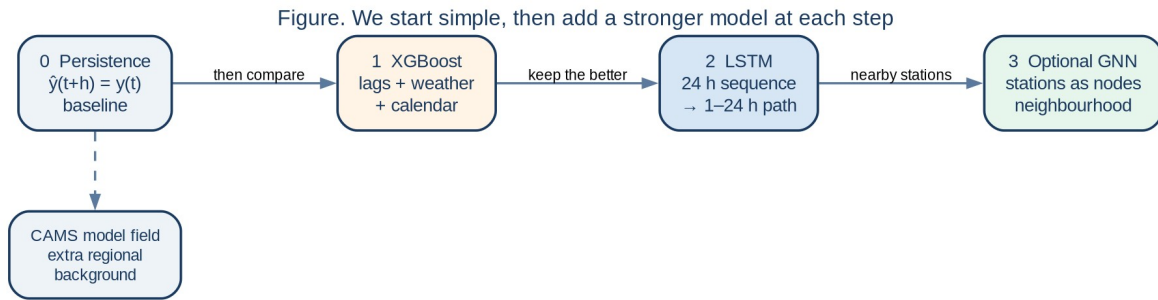


Figure 5. Model ladder. We start simple, then add XGBoost, LSTM and a station graph, and keep the better model.

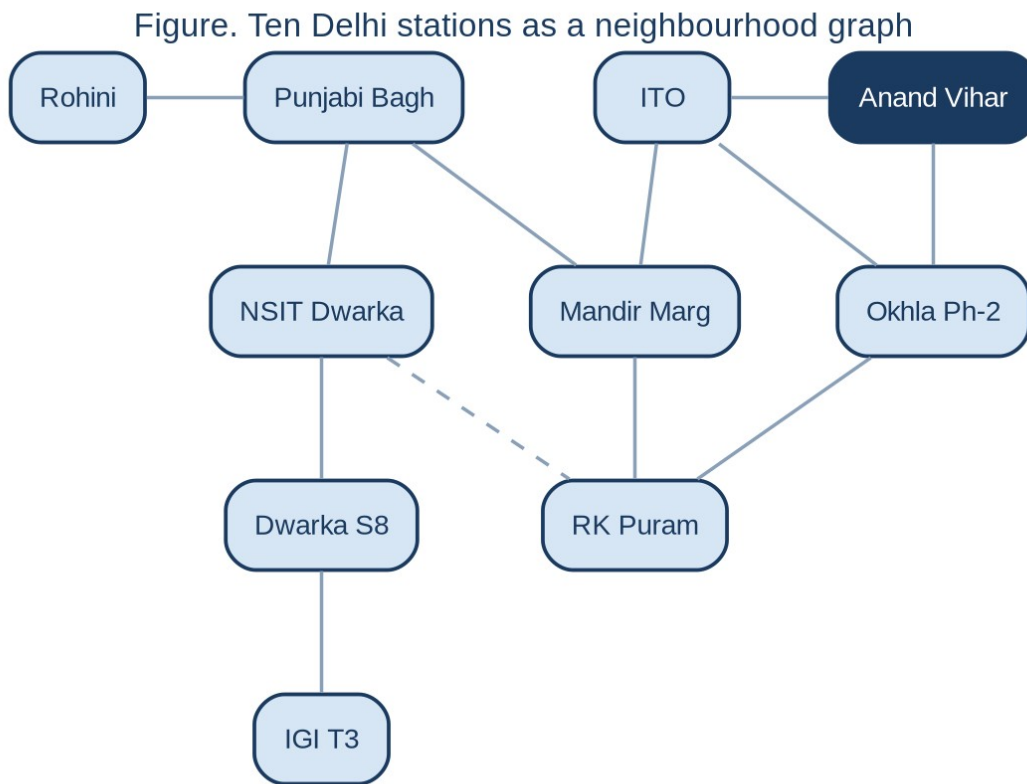


Figure 6. The same ten stations as a neighbourhood graph. Dashed edge = weaker / longer link. This is the GNN topology if Phase 6 is reached.

Every model is scored on the same frozen hold-out with the same three numbers. Persistence defines skill = 0. We keep the model that improves on that score.

Table 8. Evaluation protocol. Scores are computed on the frozen hold-out after training.

Model	MAE ($\mu\text{g}/\text{m}^3$)	RMSE ($\mu\text{g}/\text{m}^3$)	Skill vs persistence
Persistence ($t \rightarrow t+h$)	reference MAE	reference RMSE	0 by definition
XGBoost / LightGBM	primary production score	primary production score	first model we try to improve
LSTM	compared on the same split	compared on the same split	kept when it improves on boosting

GNN / spatial lag	compared on the same split	compared on the same split	kept when it improves on LSTM
CAMS field as forecast	extra comparison	extra comparison	regional background

18. Evaluation, Drift and Continuous Learning

Metrics: MAE and RMSE on PM2.5; also MAE on the derived AQI sub-index (because a 10 $\mu\text{g}/\text{m}^3$ error matters more near a category boundary). Split by season (pre-monsoon, monsoon, post-monsoon, winter) and by hour of day. Pinball loss if a later version emits intervals.

Drift. A model trained on 2024 may degrade in a later winter if emission patterns or meteorology shift. The weekly job plots rolling 28-day MAE. If MAE rises by a pre-set fraction versus the deployment baseline (for example +25 % for two consecutive weeks), the system flags that it is time to retrain. We compare the new model with the old one and keep the better one. That is continuous learning as a weekly loop.

19. Dashboard Specification

The interface is a single-city, multi-station map plus a station detail view. Minimum viable content:

- Selected station name, local time, current PM2.5, derived AQI number and category colour.
- Current PM10, NO2, temperature, humidity, wind (from the live merge).
- PM2.5 forecast chart: now, +2 h, +4 h, +6 h, +12 h, +24 h.
- One-line narrative, templated from the slope of the first three forecast hours (improve / stable / deteriorate).
- Delhi sensor map with the ten stations coloured by current category.
- Data-source badge: “CPCB via data.gov.in, weather via Open-Meteo”, plus a staleness clock.
- Last 28 days of forecast error (MAE) so users can see how the model is doing.

The interface is a map dashboard (Leaflet / MapLibre) with a FastAPI or Streamlit service in front of the forecast engine. The academic value is the forecast; the UI exists to show it.

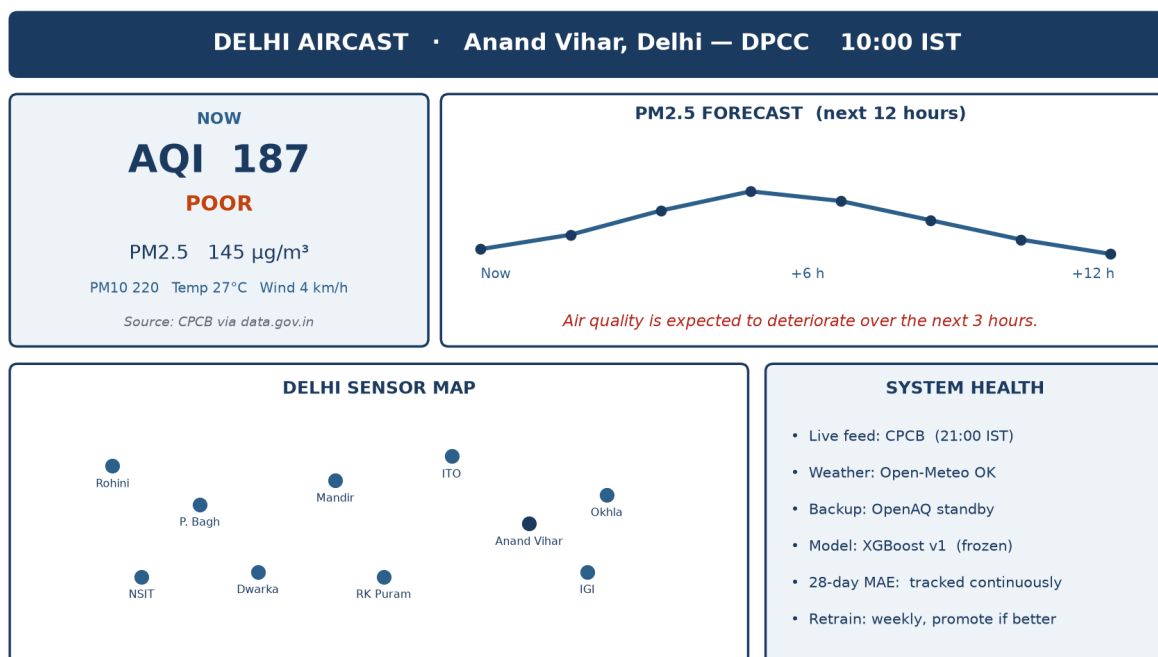


Figure 7. Target dashboard: current AQI, 12-hour PM_{2.5} path, station map, and system-health pane (source, backup, model version, 28-day MAE).

20. Free Accounts, Keys and Licensing

Table 9. Credentials. Keys stay in a local environment file for the team.

Account	Required?	Where to obtain	What it unlocks
data.gov.in API key	Yes, for live CPCB	https://data.gov.in/user/register → My Account	Full Delhi snapshot (sample key capped at 10 rows)
OpenAQ API key	Yes, for archive + backup	https://explore.openaq.org/register	v3 measurements, historical CPCB
WAQI token	Optional	https://aqicn.org/data-platform/token/	Extra live feed
Open-Meteo	No	—	Weather + CAMS already work
OpenCity CKAN	No	—	Direct CSV / package_show

Licensing summary for the report’s ethics section: CPCB observations are public environmental monitoring data accessed via OGD (NDSAP) and OpenCity (public domain metadata) / OpenAQ (provider licences, typically public domain for government monitors). Open-Meteo weather is provided for non-commercial and academic use under their terms (attribution required). WAQI is free for non-commercial use with attribution. This project is academic and non-commercial.

21. Keeping the System Reliable

21.1 How we stay online

- **Live backup.** If data.gov.in is slow, the app uses OpenAQ, then WAQI.
- **Gaps.** Missing hours are marked. The model uses the last good readings and Open-Meteo weather.
- **Units.** We check units so training stays in $\mu\text{g}/\text{m}^3$ and the dashboard stays clear.
- **More data over time.** Two full years already cover winter, monsoon, post-monsoon and Diwali. We add every new week as it arrives.

21.2 How the model learns

The model learns from history: how pollution, weather, time of day and nearby stations move together. It then gives a short-term forecast that the app turns into an Indian AQI category people can read.

22. Implementation Roadmap

Table 10. Build order, from clean data to a live weekly-improving app.

Phase	Deliverable	Depends on
0 Design freeze	This report; keys in .env	Done 7 Sep 2026
1 Data	Ten-station hourly parquet, 2024–2025, merged weather	OpenCity + Open-Meteo
2 Exploration	Diurnal, seasonal, rain, wind, Diwali, station contrast plots	Phase 1
3 Baseline	Persistence MAE/RMSE by horizon and station	Phase 1

4 ML	XGBoost/LightGBM vs baseline; feature importance	Phase 3
5 Deep learning	LSTM 24-in / 6–24-out; keep the better model	Phase 4
6 Spatial (optional)	Neighbour lags or GNN	Phase 4/5
7 Live system	Hourly job + database + map dashboard	data.gov.in + OpenAQ keys
8 Continuous learning	Weekly retrain-if-better + drift plot	Phase 7

23. Future Scope and Growth

Delhi AirCast is built to keep answering a simple question: what will PM2.5 be at this station in the next few hours? The same design also grows as we learn more and add more data.

23.1 It keeps working next year

CPCB stations such as Anand Vihar keep reporting. Open-Meteo weather goes back many years and keeps updating. We already hold the 2024–2025 station files on disk. Live updates come from data.gov.in, with OpenAQ and WAQI as extra paths. New hours go into our own store, so the project gets richer every week.

23.2 It keeps getting better

Every hour the app saves the new reading. Every week we add that week to training, compare the new model with the old one, and keep the better one. Winter, monsoon and Diwali each add new examples. Over time the forecast has more seasons to learn from.



Figure 8. Growth path: more stations, then NCR, then other cities, then alerts and a public API.

23.3 How we grow from here

Table 14. The same pipeline grows by adding stations and a few extra features.

Next step	What we add	Why it is easy
Now	10 Delhi stations, XGBoost, live dashboard	Data already checked
Deeper models	LSTM and a station graph	Same table, richer model
More of Delhi	All 39 OpenCity stations	Same files, more places
NCR	Noida, Gurugram, Ghaziabad, Faridabad	Same CPCB API, new city name
Other cities	Pune, Mumbai, Bengaluru	Same method, new station list
For people	Alerts, a public API, a school outdoor flag	The forecast is already there

24. Conclusion

Delhi AirCast is a clear college project with real Indian data already in hand. OpenCity gives two years of 15-minute CPCB station readings. Open-Meteo gives weather history and a forecast. The official data.gov.in API was live on 7 September 2026. OpenAQ and WAQI add extra live paths.

We predict PM_{2.5}, turn it into Indian AQI, and show it on a neighbourhood map. We start with a simple baseline, then XGBoost, then LSTM, then a station graph. The app updates every hour and improves every week. The same design can later cover more of Delhi, NCR and other cities. That is PS-13, ready to build and ready to grow.

25. References

- [1] Central Pollution Control Board (2014). National Air Quality Index. CPCB, Ministry of Environment, Forest and Climate Change, Government of India. https://airquality.cpcb.gov.in/ccr_docs/How_AQI_Calculated.pdf
- [2] Open Government Data Platform India. Real time Air Quality Index. Resource 3b01bcb8-0b14-4abf-b6f2-c1bfd384ba69. <https://www.data.gov.in/catalog/real-time-air-quality-index>
- [3] OpenCity / CivicDataLab. Delhi Hourly Air Quality Reports (39 stations; 2017–2023 AQI tables and 2024–25 15-minute CAAQMS extracts). <https://data.opencity.in/dataset/delhi-hourly-air-quality-reports>
- [4] OpenAQ. API v3 documentation: locations, sensors, measurements. <https://docs.openaq.org/>
- [5] OpenAQ Explorer. Anand Vihar, New Delhi – DPCC (location 235). <https://explore.openaq.org/locations/235>
- [6] Open-Meteo. Historical Weather API (ERA5 / ECMWF IFS). <https://open-meteo.com/en/docs/historical-weather-api>
- [7] Open-Meteo. Air Quality API (CAMs global / European). <https://open-meteo.com/en/docs/air-quality-api>
- [8] World Air Quality Index Project. JSON API. <https://aqicn.org/api/>
- [9] UrbanEmissions.info. CPCB daily AQI bulletins for Indian cities, 2015–2024. https://github.com/urbanemissions-info/AQI_bulletins
- [10] Vonter. india-cpcb-aqi: 15-minute CAAQMS and hourly AQI extracts. <https://github.com/Vonter/india-cpcb-aqi>
- [11] Saket Lab. vayuayan: Python clients for CPCB historical and live data. <https://github.com/saketlab/vayuayan>
- [12] Guttikunda, S. Delhi ambient PM_{2.5}, 2006–2018 (Mendeley Data). <https://doi.org/10.17632/snp7sbkb36>
- [13] Kulkarni, N. & Tawade, J. (2026). Air Quality Dataset for Delhi, 2022-08-01 to 2026-02-18 (Open-Meteo CAMS). Zenodo. <https://doi.org/10.5281/zenodo.18673773>
- [14] Kulkarni, N. & Tawade, J. (2026). Air Quality Dataset for Pune, 2022-08-01 to 2026-02-18. Zenodo. <https://doi.org/10.5281/zenodo.18677141>
- [15] Copernicus Atmosphere Monitoring Service. Global atmospheric composition forecasts. ECMWF / CAMS.
- [16] Hersbach, H. et al. (2020). The ERA5 global reanalysis. Quarterly Journal of the Royal Meteorological Society.

Appendix A. Delhi Monitoring Stations in the OpenCity Dump

The CKAN package listed 39 stations for 2017–2023 AQI tables and matching 2024–25 15-minute concentration files (package id 0dc7b9fe-9fd4-46ee-a37e-88f0bd6f6362). Phase 1 uses the 2024–25 files. CKAN package_show: https://data.opencity.in/api/3/action/package_show?id=delhi-hourly-air-quality-reports

- Alipur (DPCC)
- Anand Vihar (DPCC) — site_301 in the 2024–25 file
- Ashok Vihar (DPCC)
- Aya Nagar (IMD)
- Bawana (DPCC)
- Burari Crossing (IMD)
- Chandni Chowk (IITM)
- CRRJ Mathura Road (IMD)
- Dr. Karni Singh Shooting Range (DPCC)
- DTU (CPCB)
- Dwarka Sector 8 (DPCC)
- IGI Airport T3 (IMD)
- IHBAS Dilshad Garden (CPCB)
- ITO (CPCB)
- Jahangirpuri (DPCC)
- Jawaharlal Nehru Stadium (DPCC)
- Lodhi Road (IITM)
- Lodhi Road (IMD)
- Major Dhyani Chand National Stadium (DPCC)
- Mandir Marg (DPCC)
- Mundka (DPCC)
- Najafgarh (DPCC)
- Narela (DPCC)
- Nehru Nagar (DPCC)
- North Campus DU (IMD)
- NSIT Dwarka (CPCB)
- Okhla Phase-2 (DPCC)
- Patparganj (DPCC)
- Punjabi Bagh (DPCC)
- Pusa (DPCC)
- Pusa (IMD)
- Rohini (DPCC)
- R K Puram (DPCC)
- Shadipur (CPCB)
- Sirifort (CPCB)
- Sonia Vihar (DPCC)
- Sri Aurobindo Marg (DPCC)
- Vivek Vihar (DPCC)
- Wazirpur (DPCC)

Anand Vihar 2024–25 direct URL used in verification: <https://data.opencity.in/dataset/0dc7b9fe-9fd4-46ee-a37e-88f0bd6f6362/resource/5ef3f66f-2bb0-4593-91db-ba6e693a77f3/download/del-anand-vihar-dpcc-2024-25.csv>

Appendix B. Live Verification Log (7 September 2026)

These calls were made from a workstation in India on 7 September 2026. They show the sources we use, and how quickly they replied.

Table 11. Verification log for the sources we use.

Probe	HTTP	Time	Bytes / notes
Open-Meteo forecast weather (Delhi, 72 h)	200	0.53 s	4.1 KB; last temp 26.7 °C on 2026-09-07 23:00
Open-Meteo archive weather (2024-01-01..07)	200	0.74 s	7.5 KB; 168 hours; 00:00 temp 7.0 °C
Open-Meteo AQ current+forecast	200	0.55 s	PM2.5 103.1, PM10 113.9, US AQI 159 at 21:30
Open-Meteo AQ history (2024-01-01..07)	200	0.53 s	168 hours; PM2.5 sample 146.7, 130.4, 117.8
Open-Meteo forecast next 48 h	200	0.55 s	48 hours, 2026-09-07 00:00 → 2026-09-08 23:00
OpenCity CKAN package_show	200	0.19 s	105 KB JSON; 78 resources
OpenCity Anand Vihar 2024–25 CSV	200	3.42 s	11,812,427 bytes; 70,177 rows
data.gov.in Delhi PM2.5 (sample key)	200	~3 s	Updating 07-09-2026 21:00:00 IST
OpenAQ v3	ready	—	Free key at explore.openaq.org
WAQI personal token	ready	—	Free token at aqicn.org for an extra live feed

Appendix C. CPCB National AQI Breakpoints

Reproduced from CPCB’s National Air Quality Index methodology for the eight notified pollutants. Units are $\mu\text{g}/\text{m}^3$ except CO (mg/m^3). Averaging is 24-hour except CO and O₃ (8-hour). The location AQI is the maximum sub-index when at least three pollutants are present and one is PM_{2.5} or PM₁₀.

Table 12. CPCB health breakpoints used in the AQI engine. *O₃ very poor/severe uses 1-hour values in the official note.

Category (AQI)	PM _{2.5}	PM ₁₀	NO ₂	SO ₂	CO	O ₃	NH ₃
Good (0–50)	0–30	0–50	0–40	0–40	0–1.0	0–50	0–200
Satisfactory (51–100)	31–60	51–100	41–80	41–80	1.1–2.0	51–100	201–400
Moderately Polluted (101–200)	61–90	101–250	81–180	81–380	2.1–10	101–168	401–800
Poor (201–300)	91–120	251–350	181–280	381–800	10.1–17	169–208	801–1200
Very Poor (301–400)	121–250	351–430	281–400	801–1600	17.1–34	209–748*	1201–1800
Severe (401–500)	250+	430+	400+	1600+	34+	748+*	1800+

Linear interpolation of a sub-index I for concentration C between breakpoints B_{lo} , B_{hi} with index bounds I_{lo} , I_{hi} is $I = (I_{hi} - I_{lo}) / (B_{hi} - B_{lo}) \times (C - B_{lo}) + I_{lo}$. The AQI engine implements this exactly, with unit tests on the published examples in CPCB’s “How is AQI calculated?” note.

Appendix D. Worked Forecast Example

This is a product sketch of what the dashboard shows: recent hours, the current reading, and the next few forecast hours.

Table 13. Schematic Anand Vihar six-hour path used as the product example in the introduction.

Time (IST)	Role	PM2.5 ($\mu\text{g}/\text{m}^3$)	Weather snapshot
07:00	history	120	cool, light wind
08:00	history	131	wind easing
09:00	history	138	humidity rising
10:00	now (origin)	145	27 °C, 72 % RH, 4 km/h, 0 mm rain
11:00	forecast h=1	151	Open-Meteo forecast meteorology allowed
12:00	forecast h=2	158	
13:00	forecast h=3	163	peak in this example
14:00	forecast h=4	157	
16:00	forecast h=6	142	afternoon mixing

Derived message (template): if the slope of hours 1–3 is positive beyond a small threshold, print “Air quality is expected to deteriorate over the next 3 hours.” Convert the rolling 24-hour mean ending at each forecast hour through Appendix C to obtain the category (in the example, Poor).

Appendix E. List of Figures

All architecture figures were generated from Graphviz DOT and Matplotlib (scripts: generate_diagrams.py). Source .dot files sit beside the PNGs in the figures/ folder so they can be edited without redrawing by hand.

Table 15. Figure index.

Fig.	File	What it shows
1	fig01_architecture.png	End-to-end free-source stack
2	fig06_source_layers.png	Labels vs features vs live backups
3	fig07_station_map.png	Ten Phase-1 stations on a Delhi coordinate panel
4	fig02_two_clocks.png	Hourly forecast job vs weekly promote-if-better
5	fig03_model_ladder.png	Persistence → XGBoost → LSTM → optional GNN
6	fig04_station_graph.png	Neighbourhood graph for a later GNN
7	fig08_dashboard_wireframe.png	Target dashboard layout
8	fig05_future_growth.png	Delhi → NCR → any CAAQMS city → product layer

— *End of report* —

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